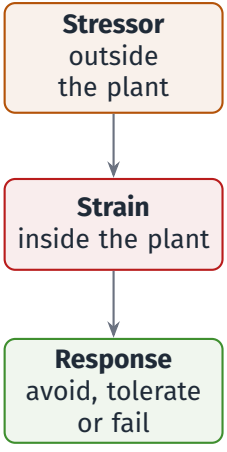
What is plant stress?

- **Stress**: any condition that limits a plant's metabolism, growth or development
- **Stressor**: the external factor (heat, drought, salt, a grazer)
- **Strain**: the change it causes inside the plant
- Grime: stress is anything that limits the rate of biomass production
- Stress is relative: a desert is not stressful to a succulent

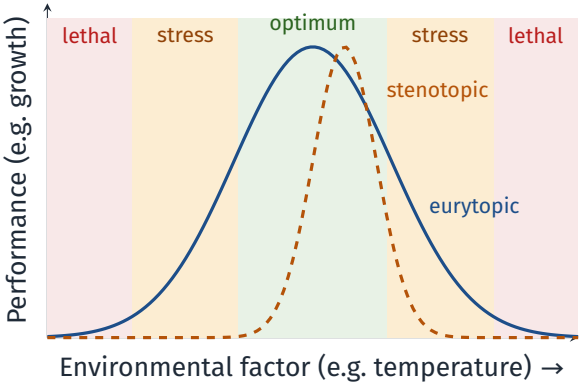


Same stressor, different strain: depends on species, tissue, life stage and history

After Lichtenthaler (1996) and Grime (1977).

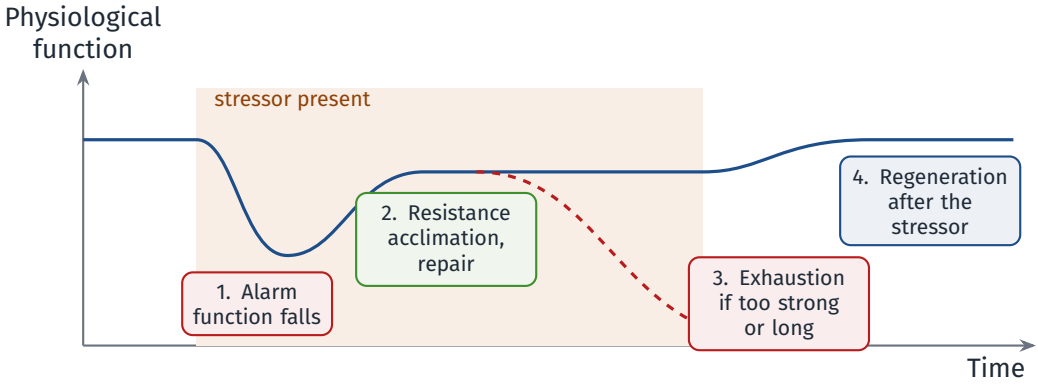
The tolerance curve

- Every plant performs best over a range of each factor: its **optimum**
- On either side: **suboptimal** zones where function declines
- Beyond the **tolerance limits**: damage and death
- Wide curve: **eurytopic** (generalist); narrow curve: **stenotopic** (specialist)
- Too little and too much can both be stressful



Schematic. Zones drawn for the wide (blue) curve.

Stress unfolds in time



Schematic, after Lichtenthaler (1996, 1998). Mild stress can harden the plant (eustress); severe stress causes lasting damage (distress).

The stressors

Abiotic

Light: too little; too much (high light, UV)
Temperature: heat, chilling, freezing
Water: drought, flooding; emersion and desiccation (intertidal)
Salinity and osmotic shock
Nutrients: deficiency or excess
Oxygen deficit (waterlogged soil)
Pollutants: ozone, heavy metals

Biotic

Herbivores and grazers
Pathogens: fungi, bacteria, viruses
Parasitic plants
Competitors: for light, water, nutrients and space
Epiphytes on seaweeds and seagrasses

Global change alters many of these at once: warmer air and sea, more frequent heatwaves and droughts, higher CO_2, ocean acidification, eutrophication and pollution. (Lists after Lichtenthaler 1998.)

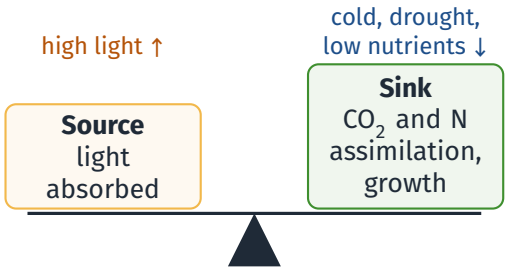
Stressors and the planetary boundaries

Boundary (Lecture 1)	Status	Stress for plants and seaweeds
Climate change	crossed	heat, drought, marine heatwaves, stronger winds and waves, sea-level rise
Ocean acidification	crossed (since 2025)	coralline algae calcify less; extra CO_2 can favour fleshy seaweeds
N and P flows	crossed	nutrient excess, eutrophication, epiphyte overgrowth, low oxygen
Freshwater change	crossed	drought, flooding, salinisation; salinity change in estuaries
Land-system change	crossed	habitat loss, fragmentation, altered microclimate
Biosphere integrity	crossed	biotic stressors: invasive species, pathogens, altered grazing
Novel entities	crossed	herbicides, heavy metals, plastics and other pollutants
Stratospheric ozone	within	UV-B stress (the ozone layer is recovering)
Atmospheric aerosols	within	less and more diffuse light; altered rainfall

Status: Richardson et al. (2023); ocean acidification from the Planetary Health Check 2025.

A common currency: the energy balance of photosynthesis

- Light capture: fast, almost insensitive to temperature
- CO_2 fixation and growth: slower; need warmth, water and nutrients
- Photon supply exceeds processing capacity, or sinks are impaired: excess energy builds up
- Sensed as **excitation pressure** on PSII

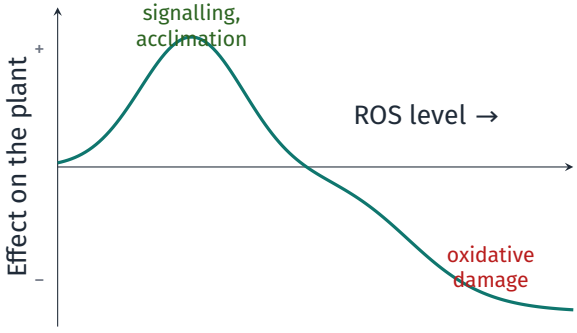


Imbalance (source > sink): PSII stays reduced; excess energy is dissipated as heat (NPQ) or forms reactive oxygen species (ROS)

After Huner et al. (1998).

Reactive oxygen species: signal and danger

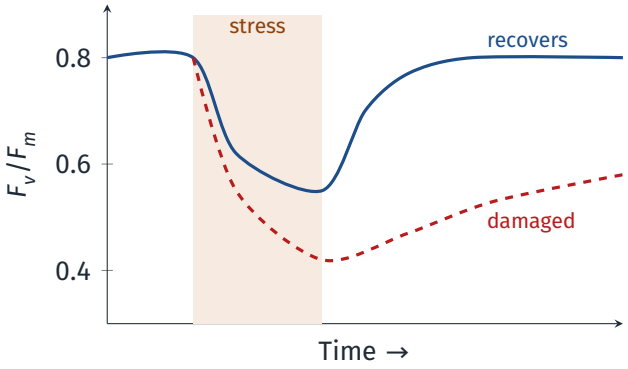
- ROS (singlet oxygen, superoxide, H_2O_2, hydroxyl radical) form in chloroplasts and mitochondria
- Almost every stress raises ROS production
- **Low levels**: signals that switch on acclimation
- **High levels**: damage to lipids, proteins and DNA
- Antioxidants (enzymes, ascorbate, glutathione, carotenoids) keep the balance



Schematic, after Apel and Hirt (2004) and Mittler et al. (2022).

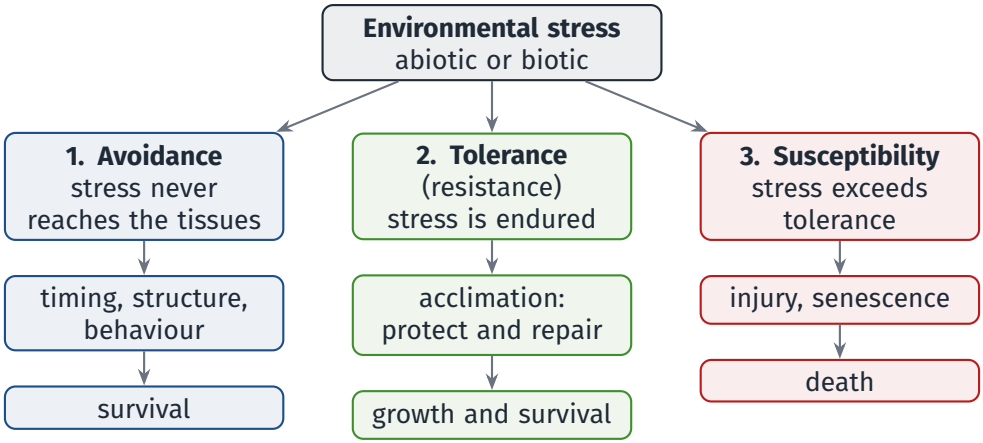
Measuring stress with chlorophyll fluorescence

- F_v/F_m: maximum quantum yield of PSII (Lecture 7b)
- Unstressed plants: about 0.8
- A fall shows that PSII is down-regulated or damaged
- **Full recovery** once the stressor is removed: photoprotection (dynamic)
- **Slow or partial recovery**: damage (chronic)
- Fast, non-destructive; works in the field and on the shore



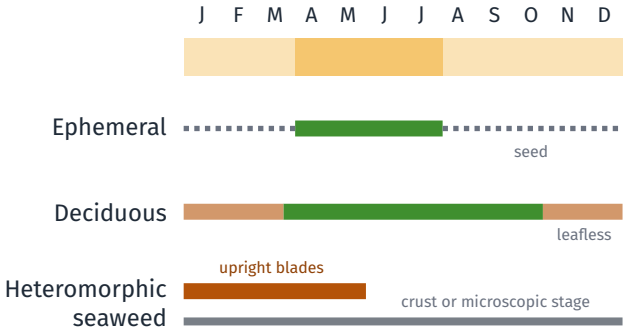
Schematic. Healthy value from Maxwell and Johnson (2000).

Three outcomes



Avoidance in time

- **Ephemerals**: live only in the favourable season; seeds survive the rest
- **Deciduous** plants drop leaves before winter or drought
- **Dormancy**: buds, bulbs, seeds, spores
- **Heteromorphic** seaweeds: upright blades alternate with a crust or microscopic stage that resists grazing



Schematic; darker band: favourable season. Seaweed example after Lubchenco and Cubit (1980).

Avoidance by structure and physiology

Land plants

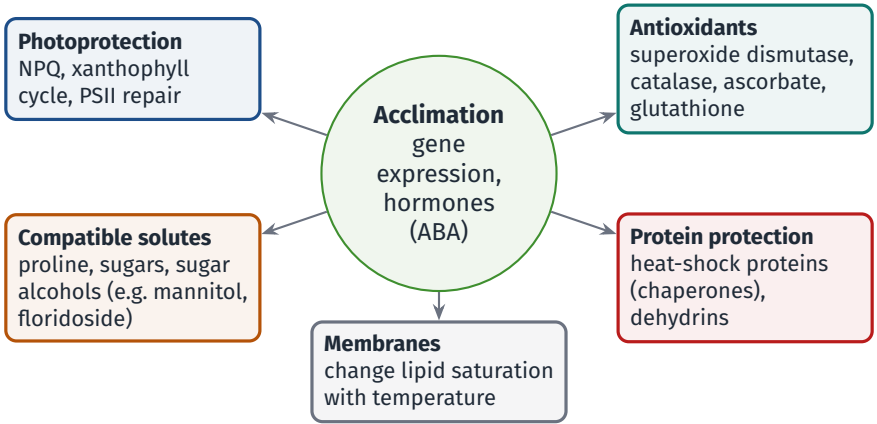
Deep roots reach groundwater
Stomata close before tissues dry
Thick cuticle, hairs, sunken stomata
Succulent tissue stores water
CAM: stomata open at night
Leaves fold or turn away from the sun

Seaweeds

Dense turfs and canopies hold water at low tide
Rock pools and crevices as refuges
Thick, mucilaginous thalli lose water slowly
Subtidal species never emerge

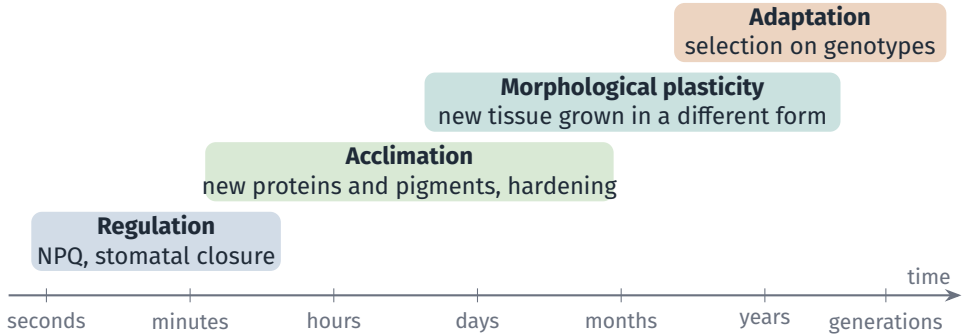
Avoidance keeps the stress outside the cells. It often costs something: closed stomata also stop CO_2 uptake.

Tolerance: the cellular toolkit



These mechanisms are switched on by the stress and switched off when it passes.

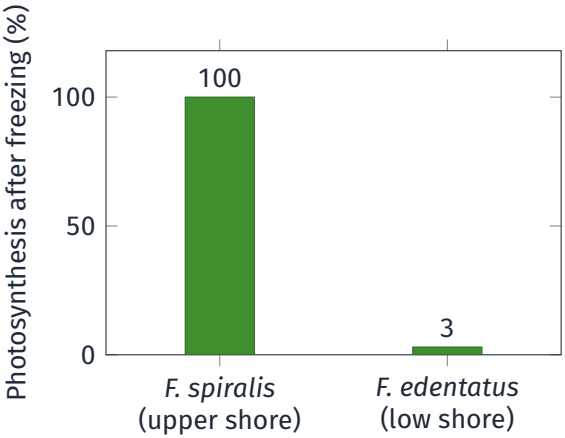
Responses on four timescales



Regulation, acclimation and morphological plasticity are changes within one genotype (phenotypic plasticity); adaptation changes the gene pool. Examples: *Ulva* pigments at depth (acclimation, Lecture 8); smaller kelps in warmer water (morphology, Wernberg et al. 2025).

Case study: stress and zonation on the shore

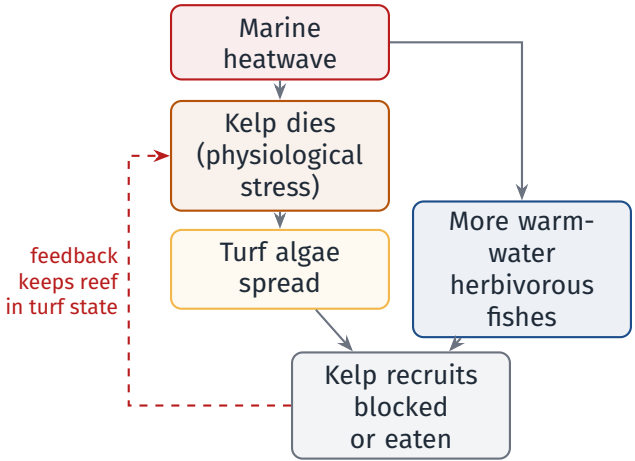
- Emersion: desiccation, heat or frost, high light, salinity change
- **Tolerance** matches shore height
- 3 h at –20 °C: upper-shore *F. spiralis* unaffected; low-shore *F. edentatus* lost 97% of photosynthesis
- Upper limits: physical stress; lower limits: competition, grazing



Davison et al. (1989); “unaffected” plotted as 100%.

Case study: a marine heatwave and a kelp forest

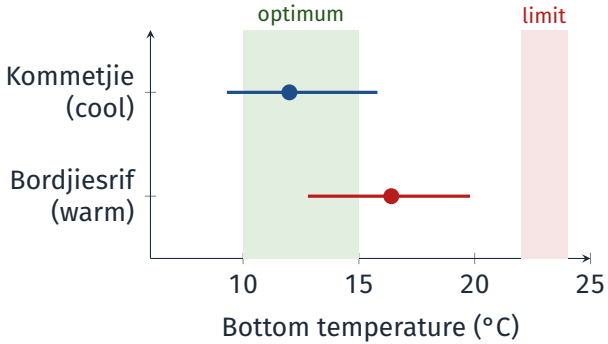
- Western Australia, 2011: >2000 km of coast, >10 weeks
- Above the net-growth limit of *Ecklonia radiata* (about 23 °C)
- About 43% of west-coast kelp died; range contracted by 100 km
- Turfs and warm-water fishes took over; no recovery



Wernberg et al. (2016); Wernberg (2021).

Case study: smaller kelps in warmer water

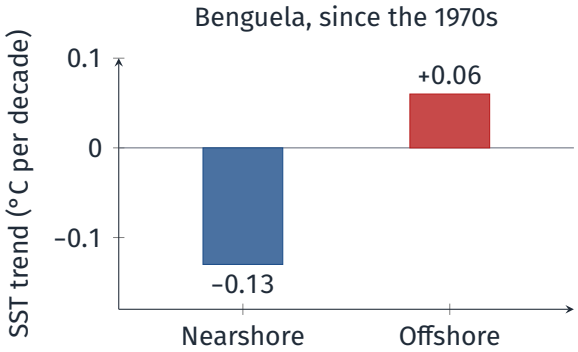
- *Laminaria pallida*, either side of the Cape Peninsula
- Warm False Bay site: shorter, thinner stipes, less biomass, no epiphytes
- Same stipe pattern in *L. hyperborea* (Norway vs France)
- **Sublethal** stress: smaller kelps, less habitat



Mean and range of temperature; optimum and survival limit for *L. pallida*. Wernberg et al. (2025).

Not every coast is warming

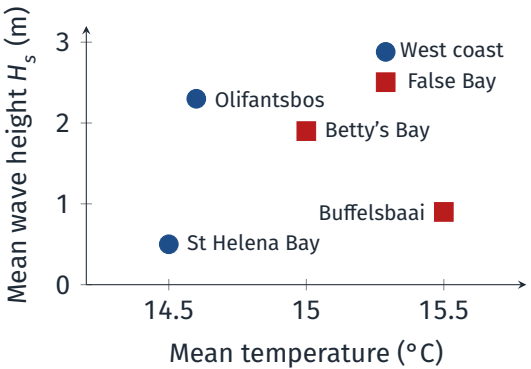
- Eastern boundary upwelling systems (EBUS): stronger upwelling-favourable winds bring up more cold water
- Southern Benguela, 1982–2009: up to −0.5 °C per decade; Agulhas Current: up to +0.55
- Eastern Agulhas Bank (summers 1983–2025): coast −0.19, offshore +0.18 °C per decade
- Explore: **www.saoce.org**



Santos et al. (2012); Rouault et al. (2010); Smit et al. (submitted).

Case study: waves and swell shape kelps

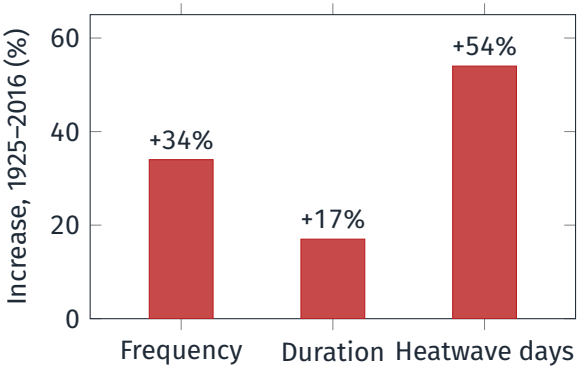
- West coast: open to Southern Ocean swell; False Bay: sheltered
- Kelp form tracked wave exposure more than temperature
- Temperature mattered more where waves were weak
- Global wave power is increasing (Reguero et al. 2019)



Annual means, end sites of each coast: Coppin et al. (2020).

Extremes are becoming more common

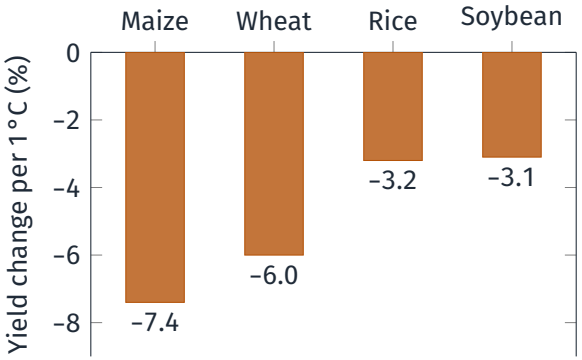
- Marine heatwaves, 1925–2016 (global average):
  - frequency +34%; duration +17%
  - heatwave days per year +54%
- Most of the trend follows the rise in mean ocean temperature
- On land, hot and dry extremes increasingly occur together



Data: Oliver et al. (2018).

Stress and food: crop yields in a warmer world

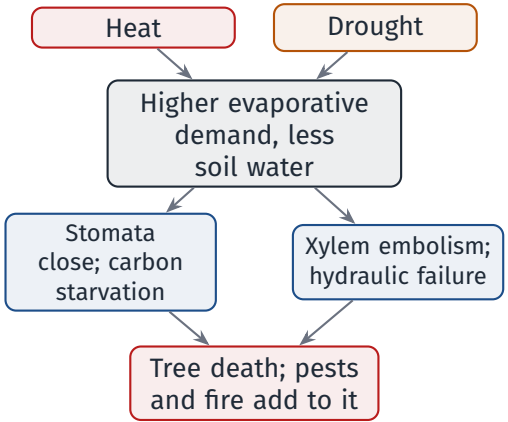
- Four crops supply about two-thirds of human calories
- Each 1 °C of global warming reduces average yields, if there is no CO_2 fertilisation, adaptation or breeding
- Effects vary strongly among regions
- Heat is more damaging when combined with drought



Data: Zhao et al. (2017); heat–drought interaction: Matiu et al. (2017).

Stress and forests: hotter droughts

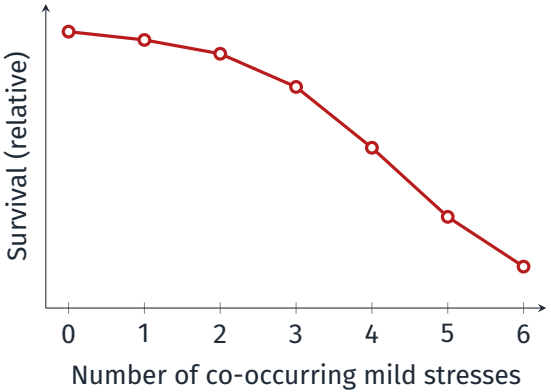
- Drought- and heat-induced tree die-off reported on every forested continent
- A global database: 154 studies, 675 locations, 1303 plots
- Mortality years were hotter **and** drier than normal: a "hotter-drought fingerprint"
- Lost trees mean lost carbon storage: a feedback to climate (Lecture 5)



Allen et al. (2010); Hammond et al. (2022).

Many small stresses add up

- In nature, stresses rarely come one at a time
- *Arabidopsis* seedlings were given up to six different stresses (including heat, salt, high light, a herbicide and cadmium), each at a mild level
- Each stress alone had little effect
- Together, growth and survival declined steeply as the number of stresses increased



Schematic of the trend reported by Zandalinas et al. (2021); not their data.

Summary

- Stress is any condition that limits metabolism, growth or development; it is defined relative to the plant
- Each plant has a tolerance curve; stress begins where function falls below the optimum
- Stress unfolds in phases: alarm, resistance, exhaustion or regeneration
- Many stresses converge on photosynthesis: absorbed energy exceeds what can be used, and ROS form
- Plants **avoid**, **tolerate** (by acclimation) or **succumb**
- Regulation (seconds), acclimation (days), morphological plasticity (months) and adaptation (generations)
- Each crossed planetary boundary is a stressor; together they add extremes and combine stresses